

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CD2-AT-1

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CIA-0620 DAA

1. Linear search - single match in middle position
given:

* Dataset: [14, 28, 35, 47, 59, 63, 71]

* Target element: 47

(i) problem interpretation:

Linear search checks each element one by one from the beginning of the list until the target element is found or the list ends. We need to find 47 in the given dataset.

(ii) step-by-step sequential search:

<u>step</u>	<u>element compared</u>	<u>Result</u>
1	14	not found
2	28	not found
3	35	not found
4	47	Found

Target 47 is found at index 3 (4th position).

(iii) no. of comparisons

14 $\rightarrow$ 1 comparison

28 $\rightarrow$ 2 comparison

35 $\rightarrow$ 3 comparison

47 $\rightarrow$ 4 comparison

total comparisons = 4

(iv) Time Complexity Analysis

Best case

* Target is the 1st element

* comparison = 1

* Time Complexity = $O(1)$

Avg case

* Target is somewhere in the middle

* Avg comparison = $n/2$

* Time Complexity = $O(n)$

Worst case

* Target is the last element or not present

* comparison = n

* Time Complexity = $O(n)$

(v) Correctness of the Result

The algorithm checks each element in order.

* 14 X

* 24 X

* 35 X

* 48 ✓

(vi) Performance Interpretation

* linear search examines elements sequentially

* since the target is in the middle, it requires

4 comparisons

(vii) note about Binary Search

* middle element = 47

* Binary search finds it in the 1st comparison

* Time complexity = $O(1)$ for this case